FOCUS 2 • PHYSICAL CHEMISTRY • 物理化学

The First Law of Thermodynamics

热力学第一定律

Energy, work, heat, enthalpy and thermochemistry

REFERENCE · 对照表

Key concepts · 重要概念

Key concepts · 中文对照 · phonetics

thermodynamics	热力学	/ˌθɜːməʊdaɪˈnæmɪks/
system	系统	/ˈsɪstəm/
surroundings	环境	/səˈraʊndɪŋz/
internal energy U	内能 U	/ɪnˈtɜːnl ˈenədʒi/
work w	功 w	/wɜːk/
heat q	热 q	/hiːt/
enthalpy H	焓 H	/ˈenθəlpi/
state function	状态函数	/steɪt ˈfʌŋkʃn/

REFERENCE · 对照表

Key concepts · 重要概念

Key concepts · 中文对照 · phonetics

heat capacity	热容	/hiːt kəˈpæsəti/
exothermic	放热	/ˌeksəʊˈθɜːmɪk/
endothermic	吸热	/ˌendəʊˈθɜːmɪk/
adiabatic	绝热	/ˌædiəˈbætɪk/
reversible	可逆	/rɪˈvɜːsəbl/
expansion work	膨胀功	/ɪkˈspænʃn wɜːk/
standard state	标准态	/ˈstændəd steɪt/
perfect gas	理想气体/完美气体	/ˈpɜːfɪkt gæs/

What we will cover

- 2A Internal energy
  内能
- 2B Enthalpy
  焓
- 2C Thermochemistry
  热化学
- 2D State functions & exact differentials
  状态函数
- 2E Adiabatic changes
  绝热变化

· The First Law is the energy book-keeping of chemistry.

Learning outcomes

- State the First Law and define internal energy U.
- Compute work and heat for expansion processes.
- Define enthalpy H and use ΔH vs ΔU correctly.
- Apply Hess's law and Kirchhoff's law.
- Recognise state functions and exact differentials.

· By the end you should be able to track energy in any change.

FOCUS 2 · WARM-UP

Where does energy come from?

Think about it · 想一想

A gas burns and heats a room. Where does that energy 'live' before it burns, and how do we count it?

Hint: think about the system (the gas) versus the surroundings (the room).

TOPIC 2A

Internal Energy

内能

➤ Why do you need to know this material?

The First Law of thermodynamics is the foundation of the discussion of the role of energy in chemistry. Wherever the generation or use of energy in physical transformations or chemical reactions is of interest, lying in the background are the concepts introduced by the First Law.

➤ What is the key idea?

The total energy of an isolated system is constant.

1. Work, heat and energy · 功、热、能量
2. Definition of internal energy · 内能定义
3. Expansion work · 膨胀功
4. Heat transactions · 热传递

The system and its surroundings

- The system is the part of the world we study.
  系统　环境
- The surroundings are everything outside, where we measure.
- Open system: matter and energy cross the boundary.
- Closed system: only energy crosses.
- Isolated system: neither crosses.

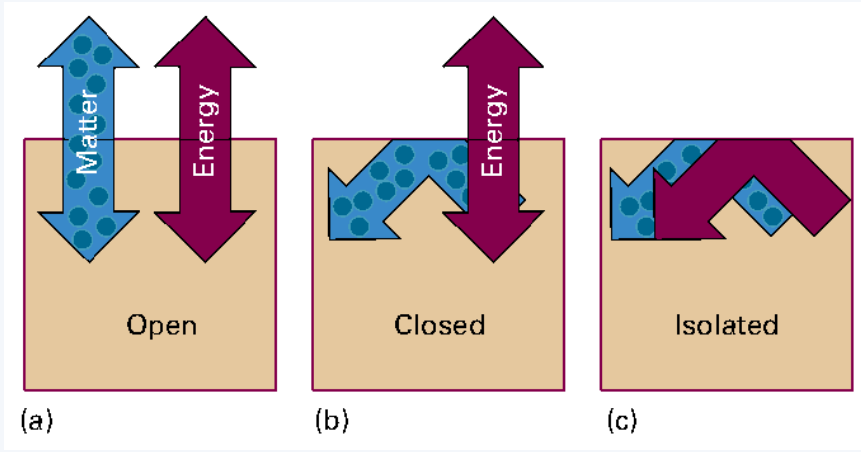


Fig 2A.1 Open, closed, isolated systems

开放、封闭、孤立

· The boundary separates the system from its surroundings.

TOPIC 2A · QUESTION

What is a boundary?

Think about it · 想一想

If you put hot coffee in a closed Thermos, is it a closed system or an isolated system?

A Thermos resists energy flow, but it is never perfectly isolated — think about why.

Work, heat, energy

- Work, w, is motion against an opposing force.
  功
- Energy, E, is the capacity to do work (1 J = 1 N·m).
  能量
- Power is the rate of energy supply (1 W = 1 J·s^{-1}).
  功率
- Kinetic energy E_k = 1/2 mv^2.
  动能
- Potential energy E_p, 势能

the energy it possessesas a result of its position. In the absence of losses, the potential energy of a stationary particle is equal to the work that had to be done on the body to bring it to its current location.

$$E_p(h) = E_p(0) + mgh$$ Gravitational potential energy

$$E_p = \frac{Q_1 Q_2}{4\pi\varepsilon r}$$ Coulomb potential energy

$$E = E_k + E_p$$ Total energy [definition]

Provided no external forces are acting on the body, its total energy is constant. This central statement of physics is known as the **law of the conservation of energy**. Potential and kinetic energy may be freely interchanged, but their sum remains constant in the absence of external influences.

· Bulk observations are explained by the molecular picture.

Exothermic and endothermic processes

- The energy of a system may be changed by means other than work itself. When the energy of a system changes as a result of a temperature difference between the system and its surroundings the energy is transferred as heat热
- Heat is a process, not an entity.

- Exothermic: energy is released to the surroundings.
  放热
- Endothermic: energy is acquired from the surroundings.
  吸热
- Example: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$, releases heat.

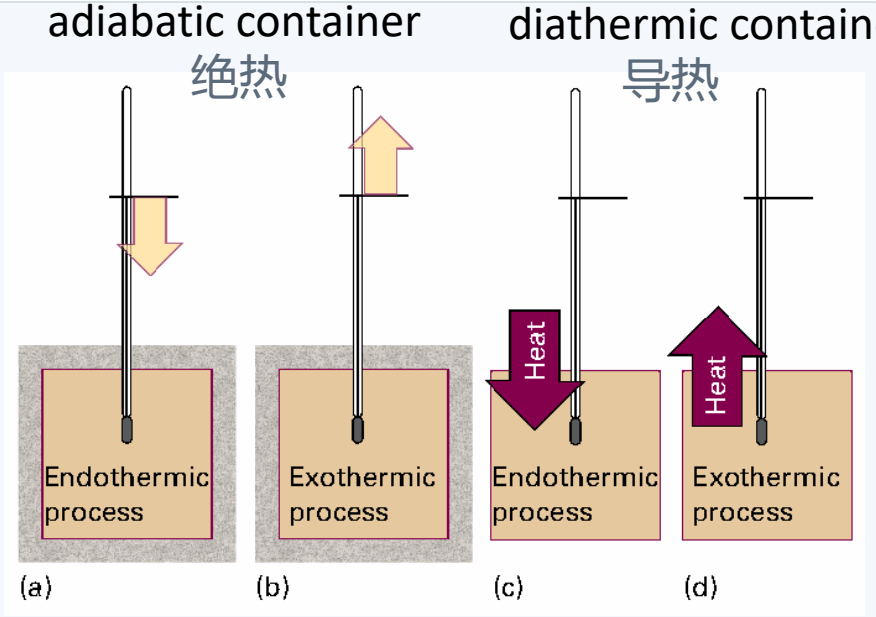


Fig 2A.2 Endothermic & exothermic

· In an adiabatic container, an exothermic process raises the temperature permanently.

Heat is disorderly; work is organised

- Thermal motion is the random motion of molecules.
- Heating uses disorderly molecular motion.
  热 → 无序
- Work uses orderly motion (a current, a raised weight).
  功 → 有序
- This is why we count heat and work separately.

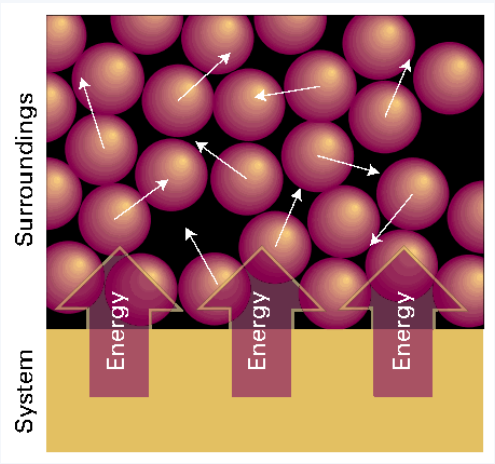


Fig 2A.3 Heat stimulates random motion

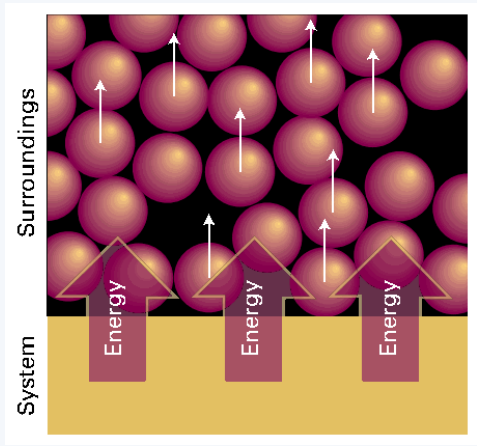


Fig 2A.4 Work uses organised motion

· The distinction matters for the First Law and for entropy later.

Definition of internal energy

Suppose a sample is divided into smaller samples.
If a property of the original sample has a value equal to the sum of its values in all the smaller samples, it is extensive. 广度性质
If a property retains the same value as in the original sample for all the smaller samples, it is intensive 强度性质

- U is the total energy of the system.
  内能 U
- It is the sum of the kinetic and potential energies of all particles.
- U is a state function — it depends only on the current state.
  状态函数
- A state function is a property that depends only on the current state of a system and is independent of its history. Physical quantities with values that do depend on the path between two states are called path functions.

	State Function	Path Function
Determined by	Only initial and final states; path-independent	Depends on the process/path
Notation	$\Delta X = X_{final} - X_{initial}$	Never write Δq or Δw. Use q, w directly
Examples	U,H,S,G,T,p,V	Heat q, Work w

· We can never measure absolute U, only changes in it.

The First Law of Thermodynamics

- U is measured in joules (1 J = 1 kg m^2 s^{-2}). The molar internal energy, U_m, is the internal energy divided by the amount of substance in a system, U_m =U/n (kJ mol^{-1}).
- Heat and work are equivalent ways of changing the internal energy of a system.

First Law $\Delta U = q + w$

热力学第一定律

Differential $dU = dq + dw$

Isolated system $\Delta U = 0$ **(energy is conserved)**

The First Law is the conservation of energy: you cannot get something for nothing.

Brief illustration 2A.2

If an electric motor produces 15 kJ of energy each second as mechanical work and loses 2 kJ as heat to the surroundings, then the change in the internal energy of the motor each second is
$\Delta U = -2\ \text{kJ} - 15\ \text{kJ} = -17\ \text{kJ}$.

Suppose that, when a spring弹簧 is wound上紧, 100 J of work is done on it but 15 J escapes to the surroundings as heat. The change in internal energy of the spring is
$\Delta U = 100\ \text{J} - 15\ \text{J} = +85\ \text{J}$.

A note on good practice Always include the sign of ΔU (and of ΔX in general), even if it is positive.

TOPIC 2A · EXAMPLE 1

Worked example: applying the First Law

Problem · 问题

A gas absorbs 500 J of heat and does 200 J of work while expanding. Find ΔU.

Step 1: Heat absorbed, so q = +500 J.
Step 2: The gas does 200 J of work, so w = −200 J.
Step 3: ΔU = q + w = 500 − 200.

Answer · 答案: ΔU = +300 J (internal energy increases)

Work done BY the system is negative.

Expansion work: a gas pushes a piston

- A gas expanding against an external pressure does work.
  膨胀功
- For a piston活塞 of area A, $dw = -p_{ex}dV$.
- The negative sign: the system loses energy on expansion.
- Total work is the integral积分 over the volume change.

$$w = -\int_{V_i}^{V_f} p_{ex}\,dV$$

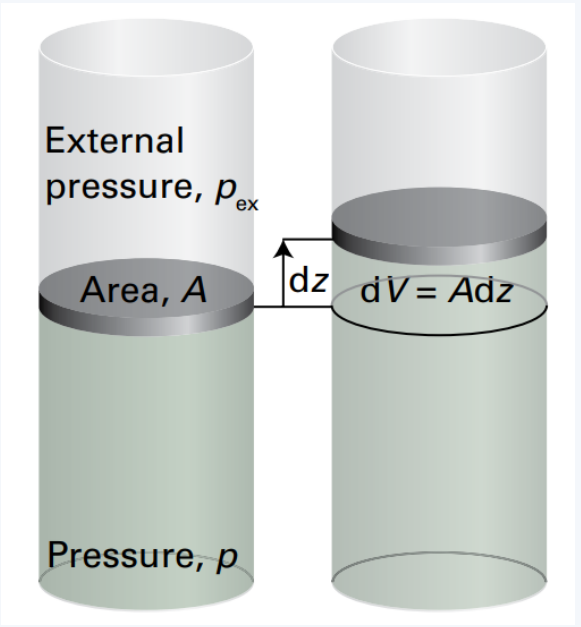


Fig 2A.5 Expansion work

· Work done BY the gas equals minus the work done ON it.

Table 2A.1 Varieties of work*

Type of work	d*w*	Comments	Units†
Expansion 膨胀功	$-p_{ex}dV$	p_{ex} is the external pressure	Pa
		dV is the change in volume	m^3
Surface expansion 表面功	$\gamma d\sigma$	γ is the surface tension	$N\ m^{-1}$
		dσ is the change in area	m^2
Extension 弹性功	fdl	f is the tension	N
		dl is the change in length	m
Electrical 电功	ϕdQ	ϕ is the electric potential	V
		dQ is the change in charge	C
	$Qd\phi$	dϕ is the potential difference	V
		Q is the charge transferred	C

* In general, the work done on a system can be expressed in the form $dw = -|F|dz$, where $|F|$ is the magnitude of a 'generalized force' and dz is a 'generalized displacement'.

† For work in joules (J). Note that 1 N m = 1 J and 1 V C = 1 J.

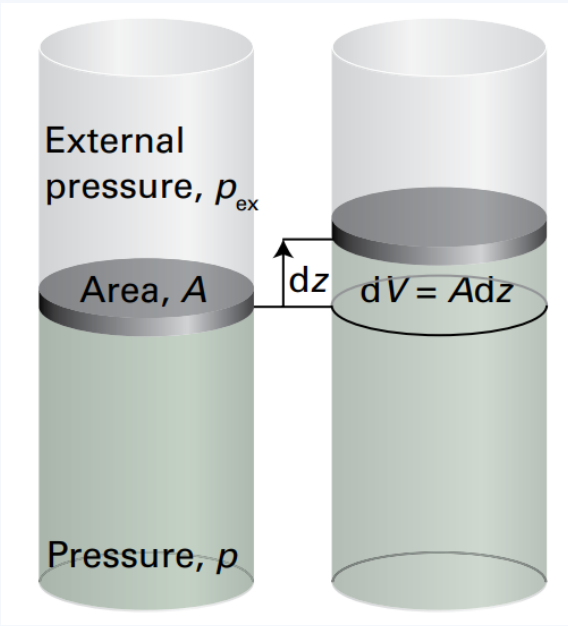


Fig 2A.5 Expansion work

Expansion against constant pressure

$$w = -\int_{V_i}^{V_f} p_{ex}\,dV$$

$$w = -p_{ex}\int_{V_i}^{V_f} dV = -p_{ex}(V_f - V_i)$$

Therefore, if the change in volume is written as $\Delta V = V_f - V_i$,

$$w = -p_{ex}\Delta V \qquad \text{Expansion work [constant external pressure]} \qquad (2A.6)$$

Free expansion is expansion against zero opposing force. It occurs when $p_{ex} = 0$. According to eqn 2A.6, in this case

$$w = 0 \qquad \text{Work of free expansion} \qquad (2A.7)$$

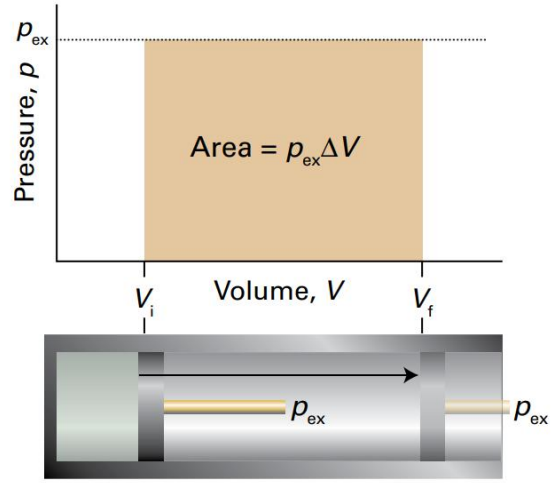


Figure 2A.6 The work done by a gas when it expands against a constant external pressure, p_{ex}, is equal to the shaded area in this example of an indicator diagram.

- Fig. 2A.6 makes use of the fact that the magnitude of an integral can be interpreted as an area. The magnitude of w, denoted |w|, is equal to the area beneath the horizontal line at p =p_{ex} lying between the initial and final volumes. A p,V-graph used to illustrate expansion work is called an indicator diagram 示功图

Work is not a state function — it depends on the path.

Example 2A.1

Calculate the work done when 50 g of iron reacts with hydrochloric acid to produce $FeCl_2(aq)$ and hydrogen in (a) a closed vessel of fixed volume, (b) an open beaker at 25°C

Collect your thoughts You need to judge the magnitude of the volume change and then to decide how the process occurs. If there is no change in volume, there is no expansion work however the process takes place. If the system expands against a constant external pressure, the work can be calculated from eqn 2A.6. A general feature of processes in which a condensed phase changes into a gas is that you can usually neglect the volume of a condensed phase relative to the volume of the gas it forms.

Example 2A.1

Calculate the work done when 50 g of iron reacts with hydrochloric acid to produce $FeCl_2(aq)$ and hydrogen in (a) a closed vessel of fixed volume, (b) an open beaker at 25°C

The solution In (a) the volume cannot change, so no expansion work is done and $w = 0$. In (b) the gas drives back the atmosphere and therefore $w = -p_{ex}\Delta V$. The initial volume can be neglected because the final volume (after the production of gas) is so much larger and $\Delta V = V_f - V_i \approx V_f = nRT/p_{ex}$, where n is the amount of H_2 produced. Therefore,

$$w = -p_{ex}\,\Delta V \approx -p_{ex} \times \frac{nRT}{p_{ex}} = -nRT$$

Because the reaction is $Fe(s) + 2\,HCl(aq) \rightarrow FeCl_2(aq) + H_2(g)$, 1 mol H_2 is generated when 1 mol Fe is consumed, and n can be taken as the amount of Fe atoms that react. Because the molar mass of Fe is 55.85 g mol^{-1}, it follows that

$$w = -\frac{50\,\text{g}}{55.85\,\text{g mol}^{-1}} \times (8.3145\,\text{J K}^{-1}\,\text{mol}^{-1}) \times (298\,\text{K})$$

$$\approx -2.2\,\text{kJ}$$

The magnitude of the external pressure does not affect the final result: the lower the pressure, the larger is the volume occupied by the gas, so the effects cancel.

Example 2A.1

Self-test 2A.1 Calculate the expansion work done when 50 g of water is electrolysed under constant pressure at 25°C.

Reaction: $\mathrm{H_2O\,(l)} \longrightarrow \mathrm{H_2\,(g)} + \frac{1}{2}\,\mathrm{O_2\,(g)}$.

1 mol liquid water yields $1.5\ \mathrm{mol}$ gaseous products. The volume of liquid water is negligible, so $\Delta V \approx V_{\mathrm{gas}}$.

Constant-pressure expansion work:

$$w = -p\Delta V \approx -\Delta n_{\mathrm{gas}} RT$$

$$n(\mathrm{H_2O}) = \frac{50\ \mathrm{g}}{18.02\ \mathrm{g\ mol^{-1}}} = 2.775\ \mathrm{mol}$$

$$\Delta n_{\mathrm{gas}} = 2.775 \times 1.5 = 4.163\ \mathrm{mol}$$

$$\begin{aligned} w &= -(4.163\ \mathrm{mol}) \times 8.314\ \mathrm{J\ mol^{-1}K^{-1}} \times 298.15\ \mathrm{K} \\ &= -10.3\ \mathrm{kJ} \end{aligned}$$

Reversible expansion可逆膨胀

- A reversible change in thermodynamics is a change that can be reversed by an infinitesimal modification of a variable.
- Reversible expansion: $p_{ex} = p$ at every step.

$$dw = -p_{ex}dV = -pdV \quad \text{Reversible expansion work} \quad (2A.8a)$$

$$w = -\int_{V_i}^{V_f} p\,dV \quad (2A.8b)$$

- Perfect gas, constant T: p = nRT/V.

$$w = -nRT \overbrace{\int_{V_i}^{V_f} \frac{dV}{V}}^{\text{Integral A.2}} = -nRT \ln \frac{V_f}{V_i} \quad \text{Work of isothermal reversible expansion [perfect gas]} \quad (2A.9)$$

Brief Illustration 2A.3

When a sample of 1.00 mol Ar, regarded here as a perfect gas, undergoes an isothermal reversible expansion at 20.0 °C from 10.0 dm^3 to 30.0 dm^3 the work done is

$$w = -(1.00\,\text{mol}) \times (8.3145\,\text{J K}^{-1}\,\text{mol}^{-1}) \times (293.2\,\text{K}) \ln \frac{30.0\,\text{dm}^3}{10.0\,\text{dm}^3}$$

$$= -2.68\,\text{kJ}$$

· Maximum work is obtained from a reversible expansion.

Reversible expansion可逆膨胀

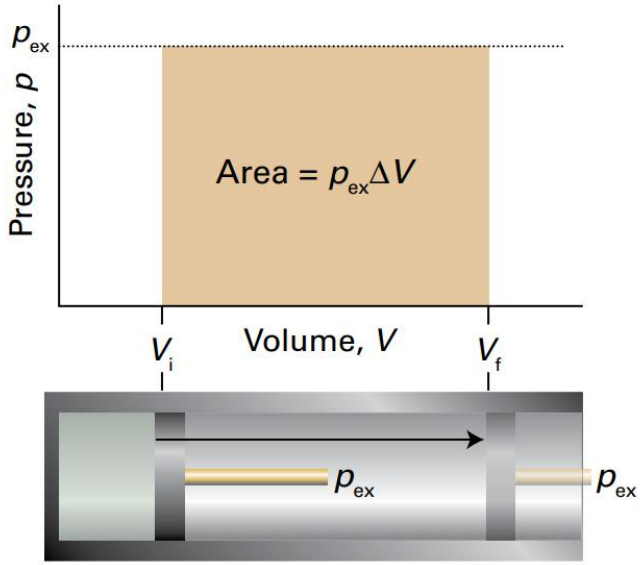


Figure 2A.6 The work done by a gas when it expands against a constant external pressure, p_{ex}, is equal to the shaded area in this example of an indicator diagram.

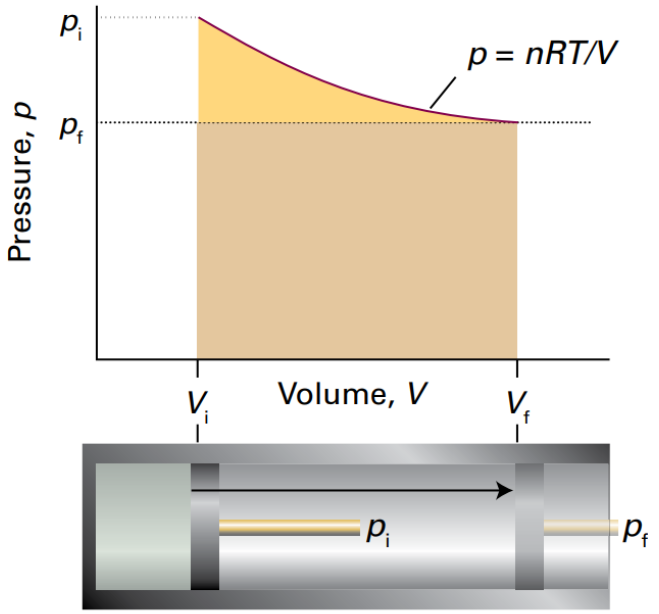


Figure 2A.7 The work done by a perfect gas when it expands reversibly and isothermally is equal to the area under the isotherm $p = nRT/V$. The work done during the irreversible expansion against the same final pressure is equal to the rectangular area shown slightly darker. Note that the reversible work done is greater than the irreversible work done.

· Maximum work is obtained from a reversible expansion.

Heat transactionsl

- In general, the change in internal energy of a system is

$$dU = dq + dw_{exp} + dw_{add}$$

- dw_{add} is work in addition to the expansion work, dwexp. For instance, dw_{add} might be the electrical work of driving a current of electrons through a circuit.
- A system kept at constant volume can do no expansion work, so $dw_{exp} = 0$. If the system is also incapable of doing any other kind of work, then $dw_{add} = 0$ too. Under these circumstances:

$$dU = dq \qquad dU = dq$$

Heat transferred at constant volume (2A.11a)

$$\overbrace{\int_i^f dU}^{U_f - U_i} = \overbrace{\int_i^f dq_V}^{q_V} \qquad \Delta U = q_V$$

· We measure q_V to find ΔU. dq is not written as Δq because q, unlike U, is not a state function.

Calorimetry量热法/量热学

- Calorimetry is the study of the transfer of energy as heat during a physical or chemical process.
- A calorimeter量热计 is a device for measuring energy transferred as heat.
- This is the adiabatic bomb calorimeter.弹式量热计
- q_V and ΔU can be determined by measuring ΔT.
- Calorimeter constant, C:

$$q = C\Delta T$$

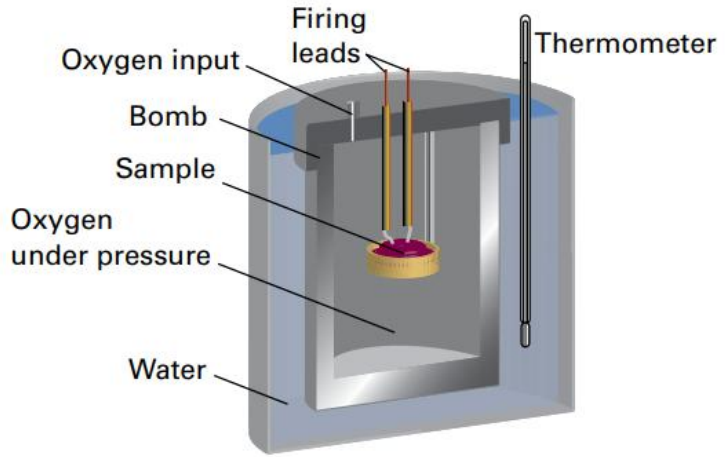


Figure 2A.8 A constant-volume bomb calorimeter. The 'bomb' is the central vessel, which is strong enough to withstand high pressures. The calorimeter is the entire assembly shown here. To ensure adiabaticity, the calorimeter is immersed in a water bath with a temperature continuously readjusted to that of the calorimeter at each stage of the combustion.

· We measure q_V to find ΔU.

Heat capacity热容

- The internal energy of a system increases when its temperature is raised. This increase depends on the conditions under which the heating takes place. Suppose the system has a constant volume. If the internal energy is plotted against temperature, a curve may be obtained.
- The slope of the tangent to the curve at any temperature is called the heat capacity of the system at that temperature. The heat capacity at constant volume is denoted C_V and is defined formally as

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V$$

Heat capacity at constant volume [definition] (2A.14)

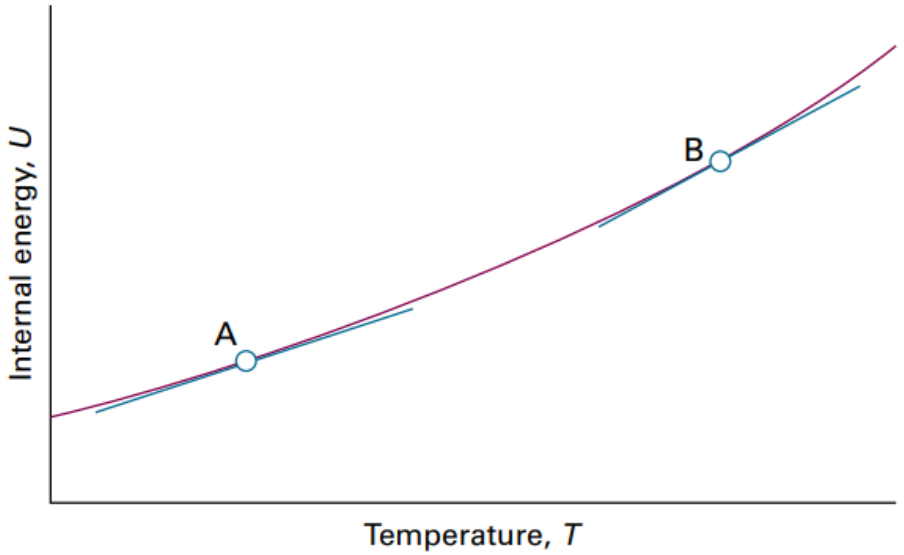


Figure 2A.9 The internal energy of a system increases as the temperature is raised; this graph shows its variation as the system is heated at constant volume. The slope of the tangent to the curve at any temperature is the heat capacity at constant volume at that temperature. Note that, for the system illustrated, the heat capacity is greater at B than at A.

Heat capacity热容

$$dU = C_V dT$$ Internal energy change on heating [constant volume] (2A.15a)

$$\Delta U = \int_{T_1}^{T_2} C_V \, dT = C_V \int_{T_1}^{T_2} dT = C_V \overbrace{(T_2 - T_1)}^{\Delta T}$$

$$\Delta U = C_V \Delta T$$ Internal energy change on heating [constant volume] (2A.15b)

$$q_V = C_V \Delta T$$ (2A.16)

- A measured quantity of energy is transferred as heat to the sample under constant volume conditions and the resulting increase in temperature is monitored.
- The ratio of the energy transferred as heat to the temperature rise it causes (qV/ΔT) is the constant-volume heat capacity of the sample.
- A large heat capacity implies that, for a given quantity of energy transferred as heat, there will be only a small increase in temperature

· We measure C_V to find ΔU.

Reversible work

Problem · 问题

1 mol of a perfect gas at 298 K expands reversibly and isothermally from 1.0 L to 10.0 L. Find the work done by the gas.

Step 1: $w = -nRT \ln(V_f/V_i)$.
Step 2: $w = -1 \times 8.314 \times 298 \times \ln(10)$.
Step 3: $\ln 10 = 2.303$, so $w = -5707$ J.

Answer · 答案: Work done by the gas = +5.71 kJ

For a reversible isothermal change of a perfect gas, $\Delta U = 0$.

Checklist of concepts

- ☐ 1. **Work** is the process of achieving motion against an opposing force.
- ☐ 2. **Energy** is the capacity to do work.
- ☐ 3. An **exothermic process** is a process that releases energy as heat.
- ☐ 4. An **endothermic process** is a process in which energy is acquired as heat.
- ☐ 5. **Heat** is the process of transferring energy as a result of a temperature difference.
- ☐ 6. In molecular terms, work is the transfer of energy that makes use of organized motion of atoms in the surroundings and heat is the transfer of energy that makes use of their disorderly motion.
- ☐ 7. **Internal energy**, the total energy of a system, is a state function.
- ☐ 8. The internal energy increases as the temperature is raised.
- ☐ 9. The **equipartition theorem** can be used to estimate the contribution to the internal energy of each classically behaving mode of motion.
- ☐ 10. The **First Law** states that the internal energy of an isolated system is constant.
- ☐ 11. Free expansion (expansion against zero pressure) does no work.
- ☐ 12. A **reversible change** is a change that can be reversed by an infinitesimal change in a variable.
- ☐ 13. To achieve **reversible expansion**, the external pressure is matched at every stage to the pressure of the system.
- ☐ 14. The energy transferred as heat at constant volume is equal to the change in internal energy of the system.
- ☐ 15. **Calorimetry** is the measurement of heat transactions.

TOPIC 2A · SELF-TEST

Check your understanding

Which statement about an isolated system is correct?

- A. Matter can cross the boundary.
- B. Energy can cross the boundary.
- C. Neither matter nor energy crosses.
- D. Only heat crosses.

Answer · 答案: C — an isolated system exchanges neither matter nor energy.

For an isolated system $\Delta U = 0$, so the First Law is pure conservation.

TOPIC 2B

Enthalpy

焓

➤ Why do you need to know this material?

The concept of enthalpy is central to many thermodynamic discussions about processes, such as physical transformations and chemical reactions taking place under conditions of constant pressure.

➤ What is the key idea?

A change in enthalpy is equal to the energy transferred as heat at constant pressure.

1. Definition H = U + pV · 定义
2. Heat at constant pressure · 恒压热
3. ΔH vs ΔU · ΔH 与 ΔU
4. Heat capacities Cv and Cp · 热容

TOPIC 2B · QUESTION

Why not just measure heat?

Think about it · 想一想

Most chemistry happens in an open beaker at constant pressure — but a gas does work as it expands. How do we account for that work?

Enthalpy is defined to absorb the pV work automatically.

Why we need enthalpy

- Most chemistry happens at constant pressure (open vessels).恒压
- Constant-pressure heat q_p is easy to measure.

- Enthalpy H = U + pV tracks q_p directly.
- H is another state function.
  状态函数

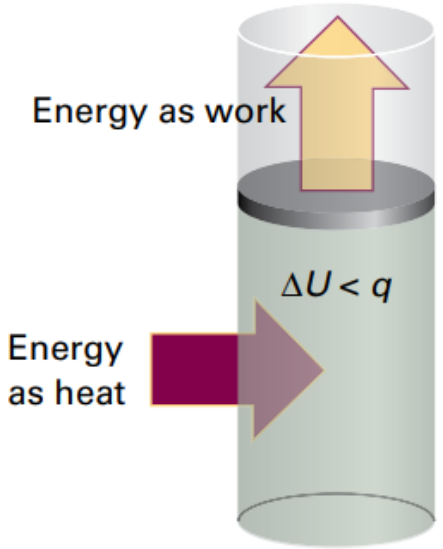


Figure 2B.1 When a system is subjected to constant pressure and is free to change its volume, some of the energy supplied as heat may escape back into the surroundings as work. In such a case, the change in internal energy is smaller than the energy supplied as heat.

· Enthalpy is U plus the energy needed to make room for the system.

Measuring heat: calorimetry

- A calorimeter measures the heat of a process.
- Flame calorimeter (constant p) gives ΔH.
  恒压量热
- Bomb calorimeter (constant V) gives ΔU.
  弹式量热计
- From the temperature change: q = C·ΔT.

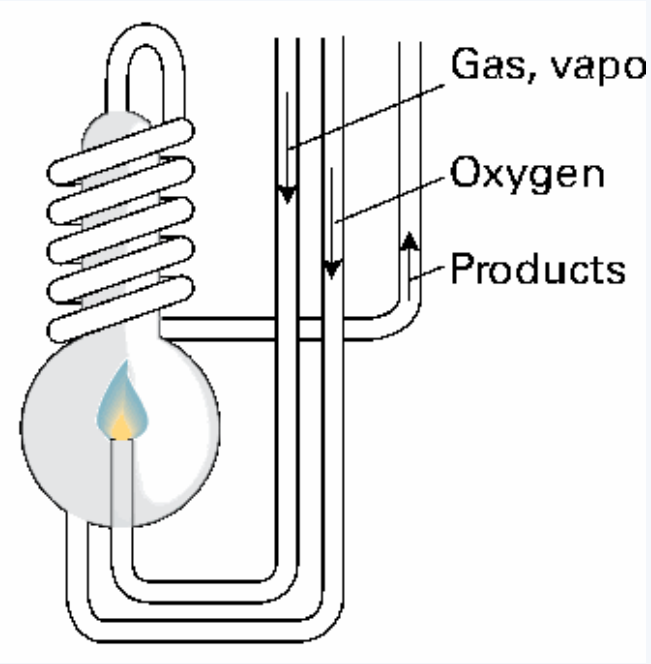


Fig 2B.2 Flame calorimeter

· Calorimetry is how reaction energies are measured experimentally.

Enthalpy and constant-pressure heat

Enthalpy $H = U + pV$
焓

At constant p $\Delta H = q_p$
恒压热

Differential $dH = dU + p\,dV + V\,dp$

$$dH = dq_p \qquad \text{(2B.2a)}$$

Heat transferred at constant pressure [infinitesimal change]

$$\Delta H = q_p \qquad \text{(2B.2b)}$$

$$\overbrace{\int_i^f dH}^{H_f - H_i} = \overbrace{\int_i^f dq}^{q_p}$$

Heat transferred at constant pressure [measurable change]

$$H + dH = (U + dU) + (p + dp)(V + dV)$$
$$= U + dU + pV + p\,dV + V\,dp + dp\,dV$$

$$H + dH = H + dU + p\,dV + V\,dp$$

$$dH = dU + p\,dV + V\,dp \qquad dU = dq + dw$$

$$dH = dq + dw + p\,dV + V\,dp \qquad dw = -p\,dV$$

$$dH = dq + V\,dp$$

At constant pressure, $dp = 0$, so

$$dH = dq \text{ (at constant pressure, no additional work)}$$

At constant pressure the heat equals the change in enthalpy.

Relating ΔH and ΔU

- The change in molar internal energy when $CaCO_3$(s) as calcite方解石converts to its polymorph aragonite文石多晶型, is +0.21 kJ mol^{-1}. Calculate the difference between the molar enthalpy and internal energy changes when the pressure is 1.0 bar. The mass densities of the polymorphs are 2.71 g cm^{-3} (calcite) and 2.93 g cm^{-3} (aragonite).

Collect your thoughts The starting point for the calculation is the relation between the enthalpy of a substance and its internal energy (eqn 2B.1). You need to express the difference between the two quantities in terms of the pressure and the difference of their molar volumes. The latter can be calculated from their molar masses, M, and their mass densities, ρ, by using $\rho = M/V_m$.

Relating ΔH and ΔU

It is usually justifiable to ignore the difference between the molar enthalpy and internal energy of condensed phases except at very high pressures when $p\Delta V_m$ is no longer negligible

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The solution The change in enthalpy when the transition occurs is

$$\begin{aligned}\Delta H_m &= H_m(\text{aragonite}) - H_m(\text{calcite}) \\ &= \{U_m(a) + pV_m(a)\} - \{U_m(c) + pV_m(c)\} \\ &= \Delta U_m + p\{V_m(a) - V_m(c)\}\end{aligned}$$

where a denotes aragonite and c calcite. It follows by substituting $V_m = M/\rho$ that

$$\Delta H_m - \Delta U_m = pM\left(\frac{1}{\rho(a)} - \frac{1}{\rho(c)}\right)$$

Substitution of the data, using $M = 100.09\,\text{g mol}^{-1}$, gives

$$\begin{aligned}\Delta H_m - \Delta U_m &= (1.0\times10^5\,\text{Pa})\times(100.09\,\text{g mol}^{-1}) \\ &\quad\times\left(\frac{1}{2.93\,\text{g cm}^{-3}} - \frac{1}{2.71\,\text{g cm}^{-3}}\right) \\ &= -2.8\times10^5\,\text{Pa cm}^3\,\text{mol}^{-1} = -0.28\,\text{Pa m}^3\,\text{mol}^{-1}\end{aligned}$$

Hence (because 1 Pa m^3 = 1 J), $\Delta H_m - \Delta U_m = -0.28\,\text{J mol}^{-1}$, which is only 0.1 per cent of the value of ΔU_m.

Self-test 2B.1

- Calculate the difference between ΔH and ΔU when 1.0 mol Sn(s, grey) of density 5.75 g cm^{-3} changes to Sn(s, white) of density 7.31 g cm^{-3} at 10.0 bar

From $H = U + pV$, at constant pressure:

$$\Delta H - \Delta U = p\Delta V$$

Molar volume $V_m = \dfrac{M}{\rho}$, $M(\mathrm{Sn}) = 118.71\ \mathrm{g\,mol^{-1}}$.

$$V_{m,\mathrm{grey}} = \frac{118.71}{5.75} = 20.645\ \mathrm{cm^3\,mol^{-1}}, \quad V_{m,\mathrm{white}} = \frac{118.71}{7.31} = 16.239\ \mathrm{cm^3\,mol^{-1}}$$

$$\Delta V = V_{\mathrm{white}} - V_{\mathrm{grey}} = -4.406\ \mathrm{cm^3} = -4.406 \times 10^{-6}\ \mathrm{m^3}$$

$$p = 10.0\ \mathrm{bar} = 1.00 \times 10^6\ \mathrm{Pa}$$

$$\Delta H - \Delta U = p\Delta V = -4.41\ \mathrm{J}$$

Relating ΔH and ΔU

- $\Delta H = \Delta U + \Delta(pV)$.
- Perfect gas at constant T: $\Delta H = \Delta U + \Delta n_g RT$.
- Δn_g is the change in moles of gas in the reaction
- If $\Delta n_g = 0$, then $\Delta H = \Delta U$.

Brief illustration 2B.2

In the reaction $2\,H_2(g) + O_2(g) \rightarrow 2\,H_2O(l)$, 3 mol of gas-phase molecules are replaced by 2 mol of liquid-phase molecules,

so $\Delta n_g = -3$ mol and $\Delta \nu_g = -3$. Therefore, at 298 K, when $RT = 2.5$ kJ mol^{-1}, the enthalpy and internal energy changes taking place in the system are related by

$$\Delta H_m - \Delta U_m = (-3) \times RT \approx -7.5\,\text{kJ mol}^{-1}$$

Note that the difference is expressed in kilojoules, not joules as in Example 2B.1. The enthalpy change is smaller than the change in internal energy because, although energy escapes from the system as heat when the reaction occurs, the system contracts as the liquid is formed, so energy is restored to it as work from the surroundings.

· This converts constant-volume data into constant-pressure data.

Heat capacity at constant pressure定压热容/恒压热容

- Cv = $(\partial U/\partial T)_V$, the constant-volume heat capacity.
  热容
- Cp = $(\partial H/\partial T)_p$, the constant-pressure heat capacity.

$$dH = C_p dT \text{ (at constant pressure)}$$

$$\Delta H = \int_{T_1}^{T_2} C_p dT = C_p \int_{T_1}^{T_2} dT = C_p \overbrace{(T_2 - T_1)}^{\Delta T}$$

$$\Delta H = C_p \Delta T \text{ (at constant pressure)}$$

$$q_p = C_p \Delta T$$

a convenient approximate empirical expression is

$$C_{p,m} = a + bT + \frac{c}{T^2}$$

- For a perfect gas, Cp − Cv = nR.
- The ratio γ = Cp/Cv is central to adiabatic work.

For a monatomic perfect gas, γ = 5/3.

· Heat capacity is the energy needed to raise the temperature by one kelvin.

TOPIC 2B · EXAMPLE 1

Worked example: raising the temperature

Problem · 问题

Heat needed to raise 2 mol of a perfect gas (Cv = 20.8 J K−1 mol−1) from 300 K to 320 K at constant volume?

Step 1: At constant volume $q_V = n \cdot Cv \cdot \Delta T$.
Step 2: $\Delta T = 20$ K.
Step 3: $q_V = 2 \times 20.8 \times 20$ J.

Answer · 答案: $q_V = +832$ J

At constant pressure you would use Cp instead of Cv.

TOPIC 2B · EXAMPLE 2

Worked example: ΔH from ΔU

Problem · 问题

For a reaction, $\Delta U = -50.0\ \text{kJ·mol}^{-1}$ at 298 K and $\Delta n_{gas} = -1$. Find ΔH.

Step 1: $\Delta H = \Delta U + \Delta n_{gas} \cdot RT$.
Step 2: $RT = 8.314 \times 298 = 2.48\ \text{kJ·mol}^{-1}$.
Step 3: $\Delta H = -50.0 + (-1)(2.48)$.

Answer · 答案: $\Delta H = -52.5\ \text{kJ·mol}^{-1}$

When gas moles decrease, ΔH is more negative than ΔU.

TOPIC 2B · SELF-TEST

Check your understanding

At constant pressure, ΔH equals:

- A. q_V
- B. q_p
- C. w
- D. ΔU − pΔV

Answer · 答案: B — at constant pressure, ΔH = q_p.

That is why chemists report enthalpies for reactions.

Checklist of concepts

- ☐ 1. Energy transferred as heat at constant pressure is equal to the change in **enthalpy** of a system.
- ☐ 2. Enthalpy changes can be measured in a constant-pressure calorimeter.
- ☐ 3. The **heat capacity at constant pressure** is equal to the slope of enthalpy with temperature.

Checklist of equations

Property	Equation	Comment	Equation number
Enthalpy	$H = U + pV$	Definition	2B.1
Heat transfer at constant pressure	$dH = dq_p$, $\Delta H = q_p$	No additional work	2B.2
Relation between ΔH and ΔU at a temperature T	$\Delta H = \Delta U + \Delta n_g RT$	Molar volumes of the participating condensed phases are negligible	2B.4
Heat capacity at constant pressure	$C_p = (\partial H/\partial T)_p$	Definition	2B.5
Relation between heat capacities	$C_p - C_V = nR$	Perfect gas	2B.9